Pradeep Yellapu

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SUMMARY

Data Analyst with **3+ years** of experience building production-grade ML pipelines, BI dashboards, and ETL workflows across finance, climate science, and enterprise IT. Fluent in Python, SQL, and Comfortable in Agile environments, working alongside Tech, Product, and Business teams. **Built GEO-AI infrastructure that cut 30+ analyst-hours to 20 minutes.** Delivered **167% engagement lift** through data-driven product decisions at a fintech startup. Currently finishing an MS in Data Science at University of Maryland (May 2026). Eligible to work in the US - no sponsorship needed immediately (**3-year OPT**).

Key outcomes: 85x faster *analyst workflows at NASA Harvest*, ***167% activation lift*** and ***40% elite plan conversion increase*** at MarketCrunch AI, and ***and 100% automation of recurring memory-threshold incidents*** at Computacenter, turning multi-source data into production pipelines and decisions that move business metrics.

WORK EXPERIENCE

NASA Harvest — **Nov 2025 - Apr 2026**
Data Science Researcher & ML Engineer — *College Park, MD*

- Built AGRA AutoPilot, a production RAG-powered GEO-AI pipeline processing SMAP soil moisture, ESI evaporative stress, and NDVI crop health satellite signals for **9 African countries**, integrated FAISS vector search, OpenAI embeddings, and GPT-4.1 mini to auto-generate structured interactive HTML bulletins; reduced monthly production from **30+ analyst-hours to ~20 minutes (85x faster)** and deployed in East African government ministries as the primary food security intelligence tool.
- Engineered multi-source geospatial ETL pipeline in Python **ingesting SMAP, ESI, NDVI, and CHIRPS rainfall datasets** across 9 African countries, automated spatial joins, feature engineering, anomaly detection, and choropleth map generation using QGIS and Matplotlib, cut analyst data preparation from days to minutes, producing analysis-ready datasets for climate resilience policymakers.
- Led the RFBS (Regional Food Balance Sheet) Technical Training Workshop in Nairobi, Kenya (March 2026), to design and deliver a **5-day hands-on curriculum** on satellite-driven crop monitoring systems and AI data pipelines for **9 senior East African government agricultural officials** across **6 nations**, enabling ministry-level independent operation of the AGRA AutoPilot system.
- Produced **2 data-driven policy briefs** synthesizing multi-signal satellite analysis (SMAP, NDVI, ESI, CHIRPS) into quantified crop risk assessments, presented at international conference in Nairobi, Kenya; findings directly informed ministerial food security policy discussions across East Africa.

MarketCrunch AI — **Jun 2025 - Sep 2025**
Data Analyst & Product Analytics Intern — *San Francisco, CA (Remote)*

- Designed and maintained SQL-based BI dashboards tracking user engagement funnels, feature adoption, and conversion metrics for **500+ enterprise users** in an Agile sprint environment; reduced reporting errors by 15% and directly informed **2 feature launches** that increased elite plan **conversion by 40%.**
- Applied cohort analysis, segmentation algorithms, and A/B testing to clickstream and session data to identify drop-off points in onboarding flow, drove **167% lift in new user activation rate** through data-driven product and behavioral analysis.

Hack4Impact — **Nov 2024 - May 2025**
Data Analyst — *College Park, MD*

- Built Power BI dashboards and SQL queries to track applicant funnel metrics across a **33-step nonprofit recruitment process** handling **750+ annual applications**, **cut recruiter review time by 80%** and improved pipeline visibility for leadership through self-service reporting framework.
- Conducted exploratory data analysis using Excel and Jupyter Notebooks to identify bottlenecks in application flow; findings directly enabled team to streamline process and reduce drop-off at critical stages.

Computacenter — **Jan 2023 - Jul 2024**
Technical Analyst — *Bengaluru, IN*

- Analyzed **~100 recurring memory-threshold ServiceNow tickets** to identify a systemic remediation gap; engineered a Python + PowerShell pipeline that detects alerts, executes RDP-based RAMMAP clearance, and logs resolution outcomes, cutting mean incident **resolution time by 40%** and eliminating manual intervention for **100% of memory threshold incidents**.
- Built IT operations BI dashboards aggregating ServiceNow ticket data, performance metrics, and resolution logs via SQL and Selenium; designed ETL workflows in Python and MySQL on AWS to enforce data governance standards and enable proactive capacity planning.

My Equation — **May 2022 - Jun 2023**
Market Research Analyst & Data Analyst — *Ahmedabad, IN*

- Conducted statistical analysis of market data using Python and R, performing regression analysis, sentiment analysis, and competitive benchmarking; findings **drove 35% revenue growth** through data-informed rebranding strategy.

PROJECTS

Transurban Express Lanes Mixed-Methods Data Analysis **Sep 2024 - Dec 2024**
University of Maryland *College Park, MD*

- Collected and analyzed quantitative survey data from **127 respondents** and qualitative data from **8 semi-structured interviews** to evaluate toll payment system usability; applied statistical analysis and SUS scoring to quantify usability gaps across wayfinding, pricing transparency, and account management workflows.
- Synthesized multi-source datasets using thematic analysis and affinity mapping; produced data-driven recommendations for signage redesign, real-time pricing display specs, and mobile app feature prioritization backed by SUS scores and behavioral metrics.

INFO Faculty Data Analysis & Dashboard Design **Jan 2025 - May 2025**
University of Maryland *College Park, MD*

- Collected and analysed faculty workload data through structured interviews and observational methods; applied affinity mapping to identify recurring inefficiency patterns across **4 key task categories**, enabling data-backed redesign of workload tracking system.
- Synthesized findings into journey maps and task flow metrics, proposed dashboard improvements that improved task ownership clarity and reduced reporting time for UMD INFO department faculty.

SKILLS

- **Programming**: Python, SQL, R, Spark, PostgreSQL, MySQL, JavaScript
- **Data Analysis & ML**: Statistical Analysis, A/B Testing, Hypothesis Testing, Regression, Clustering, Classification, Time Series Analysis, NLP, Feature Engineering, Cohort Analysis, Model Evaluation
- **Data Visualization & BI**: Tableau, Power BI, Matplotlib, Seaborn, Plotly, QGIS, Google Analytics
- **Data Engineering & Cloud:** ETL Pipelines, Data Modeling, AWS, GCP, Snowflake, Docker, API Integration, Selenium, FAISS, OpenAI API
- **Data Governance & Quality:** Data Lineage, Metadata Management, Data Quality Management, Informatica DQ concepts, WCAG/Section 508 compliance
- **Data Engineering**: Agile/Scrum, Git/GitHub, Jupyter, Excel (Advanced), Notion, Jira, PowerShell

EDUCATION

University of Maryland **Aug 2024 - May 2026**
Master of Science, Data Science *College Park, MD, USA*

- GPA: **3.85**

SRM Institute of Science & Technology **Jun 2019 - May 2023**
Bachelor of Technology, Computer Science & Business Systems *Tech Park, India*

- GPA: **3.70**

ACHIEVEMENTS

- **Co-authored 2 policy briefs** presented at the AGRA Regional Food Balance Sheet International Conference, Nairobi, Kenya - geospatial data visualizations adopted by government ministries across **9 East African nations.**
- **Creative Award of the Year** - Computacenter, 2024
- **Microsoft MLSA –** Vinci Di UI Design Challenge, **Winner - 2019**

CERTIFICATIONS

DHS Certified Trusted Tester for Web Accessibility - WCAG evaluation (Section 508) **March 2026**
CITI Program Certification - Social & Behavioral Research: ID: 67782636 **Feb. 2025**
Foundations of User Experience (UX) Design by Google: ID: 9HTB2SYBKWX9 **March 2022**